

Assignment -1

1. The three normative ethics traditions and their core questions are:

Deontology (Duty Ethics) — *“What is my duty?”*

Consequentialism (Utilitarianism) — *“What action produces the best consequences?”*

Virtue Ethics — *“What would a good/virtuous person do?”*

2. Any four features that make AI ethics distinct from “ethics as usual” are:

Scale and impact – AI systems can affect millions of people quickly and across different countries.

Autonomy and unpredictability – AI can make or influence decisions with limited direct human control, sometimes producing unexpected outcomes.

Opacity (“Black Box” problem) – It can be difficult to understand how some AI systems arrive at their decisions.

Responsibility and accountability – It can be unclear who is responsible when an AI system causes harm—the developer, company, user, or the system itself.

3. The four layers of a sociotechnical system are:

Technical layer — Example: AI algorithm/model

Individual/Human layer — Example: Users or employees operating the AI system

Organizational layer — Example: Company policies and management

Societal/Institutional layer — Example: Government laws and regulations

4. The three machine learning paradigms are:

Supervised Learning — Learns from labeled data.

Example: Email spam detection.

Unsupervised Learning — Finds patterns or groups in unlabeled data.

Example: Customer segmentation.

Reinforcement Learning — Learns through rewards and penalties by interacting with an environment.

Example: Training a robot to navigate.

5. In the self-driving car scenario, the car must choose between swerving into a wall or continuing straight.

Consequentialist:

The car should choose the action that produces the best overall outcome (e.g., the least harm or fewest deaths).

Focus: Consequences of the action.

Deontologist:

The car should follow the relevant moral duty/rule, such as not intentionally harming an innocent person. It may therefore continue straight if swerving would deliberately cause harm to someone in the car.

Focus: Duties and moral rules.

Virtue Ethicist:

The car should act in a way that reflects good moral character, such as courage, care, responsibility, and fairness. It would choose the action a virtuous decision-maker would consider morally appropriate.

Focus: Character and virtues.

Why do the answers differ?

They use different standards for deciding what is morally right:

- a. Consequentialism → What gives the best result?
- b. Deontology → What is the right duty/rule?
- c. Virtue ethics → What would a virtuous person do?

The three approaches can give different answers because consequentialism judges actions by outcomes, deontology, by duties/rules, and virtue ethics by character and virtues.

6. The statement “most ML systems are literally consequentialist by design” means that machine-learning systems usually choose actions by optimizing an objective function—a numerical measure of the desired outcome.

Objective function: The system is trained to maximize or minimize a particular outcome, such as accuracy, profit, reward, or prediction performance.

Therefore, the system effectively asks: “Which choice gives the best measurable result?” This is similar to consequentialist ethics, which judges an action mainly by its consequences.

Why add a deontological hard constraint?

An optimisation process may choose an outcome that scores well overall but violates an important moral rule.

For example, if a hiring AI is optimized only for prediction accuracy, it might discover that using race improves its performance. A consequentialist optimisation could therefore select race as an input if it improves the objective.

A deontological hard constraint—such as “never use race as an input”—prevents that action regardless of whether it improves the objective.

Key distinction

Objective function: “Choose what produces the best score/outcome.”

Hard constraint: “Some actions are forbidden, even if they would improve the score.”

ML optimisation is consequentialist-like because it selects actions based on their measurable outcomes. Deontological constraints are added to ensure that certain morally unacceptable actions remain prohibited even when optimization would otherwise favour them.

7. A purely technical audit may show that the resume-screening algorithm is mathematically unbiased—for example, it may satisfy a statistical fairness measure. However, this can miss the fact that the training data itself contains historical discrimination.

From a sociotechnical systems perspective:

Technical layer:

The algorithm may be functioning correctly and may appear mathematically unbiased. The problem is that it learns patterns from biased historical data.

Organizational layer:

The company's past hiring practices and decisions created the biased training data. If the company historically favoured one demographic, the data reflects those decisions.

Societal/institutional layer:

Broader social inequalities and discrimination may have influenced the historical hiring patterns and therefore entered the dataset.

Why the technical audit misses it

A technical audit mainly examines the model, its code, inputs, outputs, and statistical performance. It may not ask where the training data came from or why those patterns exist.

Key point:

“Unbiased algorithm + biased historical data = potentially biased system.”

The bias is not necessarily in the mathematical algorithm itself. It originates primarily in the organizational layer (historical hiring practices) and may also reflect the societal/institutional layer (existing social inequalities). This is why AI ethics requires auditing the whole sociotechnical system, not just the technical model.

8. The “problem of many hands” means that responsibility for an AI system is spread across many people and organizations, so it becomes difficult to identify one person as responsible for the harm.

In the hiring-tool example, at least these parties could share responsibility:

AI developers/data scientists — They designed, trained, tested, and deployed the model and may have failed to detect discriminatory patterns.

The company/employer — It chose to use the AI system for hiring and is responsible for how it is deployed and monitored.

Data providers or data-curators — If biased or historically discriminatory hiring data was used, those responsible for collecting and preparing the data contributed to the problem.

Management/decision-makers — They may have approved the system without adequate fairness testing, oversight, or safeguards.

Why AI makes responsibility harder to assign

With a single human decision-maker, it is relatively straightforward to identify who made the discriminatory decision. With AI, the harmful outcome can result from a chain of decisions involving developers, data providers, managers, vendors, and users.

Moreover, the system may reject thousands of applications automatically and at scale, without any individual making each rejection consciously.

AI creates a diffusion of responsibility because many actors contribute to the design, data, deployment, and use of the system. Unlike a single human error, an AI harm can emerge from the interaction of the whole system, making it difficult to determine who is ultimately accountable.

9. LLMs hallucinate because they are primarily trained to model statistical patterns in language, not to possess knowledge or beliefs in the same way humans do.

Statistical language modelling:

An LLM predicts what words/tokens are most likely to come next based on patterns learned from its training data. It does not inherently check whether the generated statement is factually true.

No genuine beliefs or understanding:

The model can produce a statement that *sounds like a fact* even when the information is missing, uncertain, or incorrect. Thus, it may generate a plausible but false answer—this is called a hallucination.

Why fluency creates “false trust”

LLMs produce answers that are often grammatically correct, coherent, confident, and human-like. This fluency can make users assume that the information is reliable even when it is wrong.

A confidently wrong answer is therefore more dangerous than an obviously wrong answer because:

Users are less likely to question or verify it.

It can be mistaken for authoritative information.

The false information may be acted upon or passed to others.

LLMs predict plausible language rather than guaranteeing truth. Their fluency can create false trust, causing users to accept incorrect information. Therefore, a confidently wrong answer can cause more harm than an obviously incorrect answer because it is more likely to be believed and acted upon.

10. Example: Biased AI hiring model

Using the capability-to-harm mapping, a biased hiring AI can be classified as a discrimination/bias harm. AI can automate and scale this harm by applying the same biased decision-making to a very large number of applicants.

Level of harm: Group/Societal-level

The worst effects are best understood as group/societal-level harm, because the system does not merely harm one individual. If it systematically rejects members of a protected group, it can create unequal employment opportunities for the whole group.

Individual-level: One qualified applicant is unfairly rejected.

Group/societal-level: Many members of the same demographic are repeatedly disadvantaged.

Systemic-level: If such biased AI becomes widespread across many organizations and reinforces existing inequalities, it can contribute to broader structural inequality.

A biased hiring model represents discrimination amplified by AI's ability to automate decisions at scale. Its primary harm is group/societal-level because it systematically disadvantages a protected group, although repeated and widespread use could also produce systemic-level harm.